



Environmental Requirements and Responsibilities



SEAPOWERTHROUGHENGINEERING

TOPIC LEARNING OBJECTIVES

Upon successful completion of this topic, the student will be able to:

- 1. Identify key Federal and DoD policies governing environmental, safety and occupational health issues.
- 2. Recognize the requirement for effective safety and health programs for every defense acquisition program.
- 3. Identify the information required for a milestone review regarding environmental, safety and health issues.
- 4. Identify the scope of the major Federal environmental regulations.
 - a. Identify the National Environmental Policy Act (NEPA) process and identify all possible outcomes of the process for a major federal action.

STUDENT PREPARATION

Student Support Material

- 1. None

Primary References

- 1. OPNAVINST 5090.1, Environmental and Natural Resources Program (SEP 2019)*being updated
- 2. OPNAV M-5090.1, Environmental Readiness Program Manual (JUN 2021)* being updated



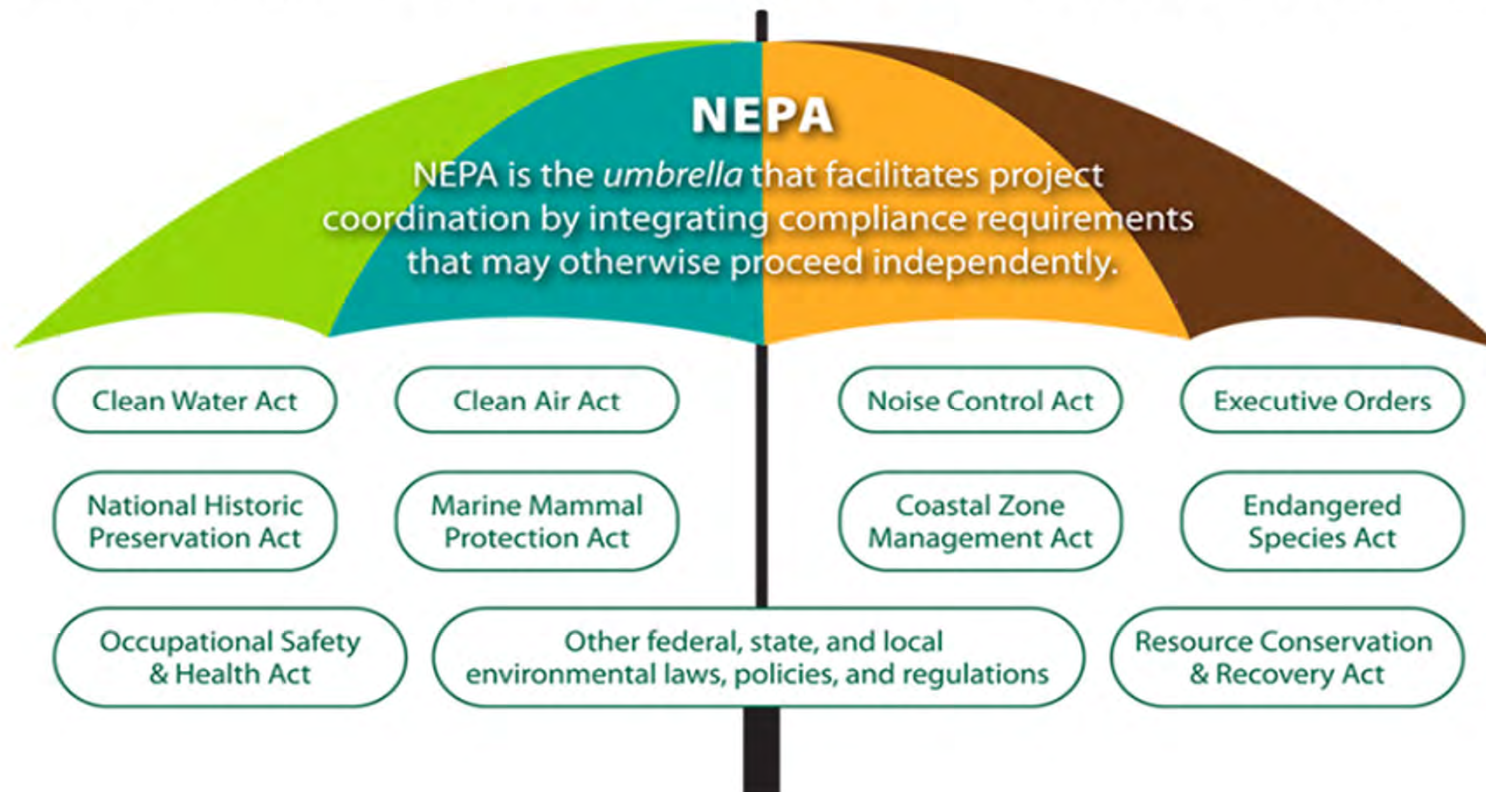
Overview

- National Environmental Policy Act (NEPA)
- ESOH Laws and Executive Orders
- DoW Policies and Requirements



NEPA is Procedural

- Established action-forcing procedures
- Federal agencies must *consider* the environmental effects of their actions
- Environmental considerations must be part of decision-making





National Environmental Policy Act (NEPA)

- For “any major federal action significantly affecting the quality of the human environment,” NEPA requires the DoW to:
 - Consider the environment in its decisions
 - Inform and involve the public in that process
 - Seek less environmentally damaging ways to accomplish the mission or operation
 - Support informed decisions with quality documents
 - PMs should design the system in such a way as to minimize any negative impact on human health and the environment
- DoW policy requires the PM of each systems acquisition program to prepare a NEPA Compliance Schedule

The NEPA Compliance Schedule is required as part of the PESHE, and should be integrated into the overall program schedule



What's New with NEPA?

- DOD NEPA Implementation Update
- EO 14154 - *Unleashing American Energy*
 - Directed CEQ to:
 - Rescind NEPA regulations – 40 CFR part 1500
 - Provide new implementation guidance to agencies.
 - CEQ Action:
 - Rescinded CEQ's NEPA regulations* effective April 11, 2025
 - Agencies instructed to:
 - Repeal existing CFR NEPA rules
 - Develop agency specific NEPA procedures



What's New with NEPA? (cont.)

- DOD NEPA Implementation
 - Developed and released new NEPA procedures, aligned with CEQ's updated guidance on June 30, 2025, for:
 - All DoW Components and US Army Corps of Engineers Civil Works
 - Purpose & Scope:
 - Establish standardized NEPA compliance procedures
 - Ensure consistency across DoW
 - Provide clear guidance for implementation & documentation



What's New with NEPA? (cont.)

- Repeal of Component NEPA regulations
 - As of June 30, 2025, all DoW Component NEPA regs in CFR were repealed
 - Replaced by enterprise-wide DOD NEPA procedures
 - New procedures:
 - No substantive changes to NEPA review process
 - Maintain continuity in NEPA compliance:
 - DOD NEPA Implementing Procedures
 - DOD NEPA Implementing Procedures: Appendix A
 - DoN will develop instruction/guidance/manual supplementing DoW procedures - *TBD timeline



DON NEPA Statement

- The Department of the Navy is committed to continued compliance with the statutory requirements of the National Environmental Policy Act. This commitment was underscored by the Department of Defense's recent issuance of DoD-wide Implementing Procedures for NEPA, which are detailed and comprehensive. DON has rescinded our previous regulations but will continue complying with the statutory requirements of NEPA, including by identifying and considering relevant environmental information to ensure informed decision making. Importantly, we intend to remain transparent and continue to communicate with stakeholders throughout the NEPA process.



When NEPA review is NOT required

- DOD NEPA review is not required if:
 - No final agency action is involved
 - The action is exempted by law
 - NEPA conflicts with other legal requirements
 - Congress limits DoW discretion by statute
 - Another law satisfies NEPA requirements
 - The action is not a major Federal action
 - Federal involvement or funding is minimal
 - Issuance of DOD NEPA procedures itself

- DoW focuses on the specific action or project at hand



Determining the level of NEPA Review

- Process for deciding NEPA review level:
 - Check for existing categorical exclusions (CATEXs):
 - Apply DoW's or another agency's CATEX if it fits the proposed action
 - Consider creating or revising CATEX if needed
 - If no CATEX applies:
 - Evaluate the action's reasonably foreseeable effects
 - If effects are not likely significant or unknown, prepare an Environmental Assessment (EA)
 - If effects are likely significant, prepare an Environmental Impact Statement (EIS)



DoW as Single Federal Agency for NEPA CATEX

- Key Point:
 - The Department of War (DoW), including all its components, is considered a single “Federal Agency” under NEPA’s CATEX provisions
 - This applies regardless of which DoW component originally established or adopted the CATEX

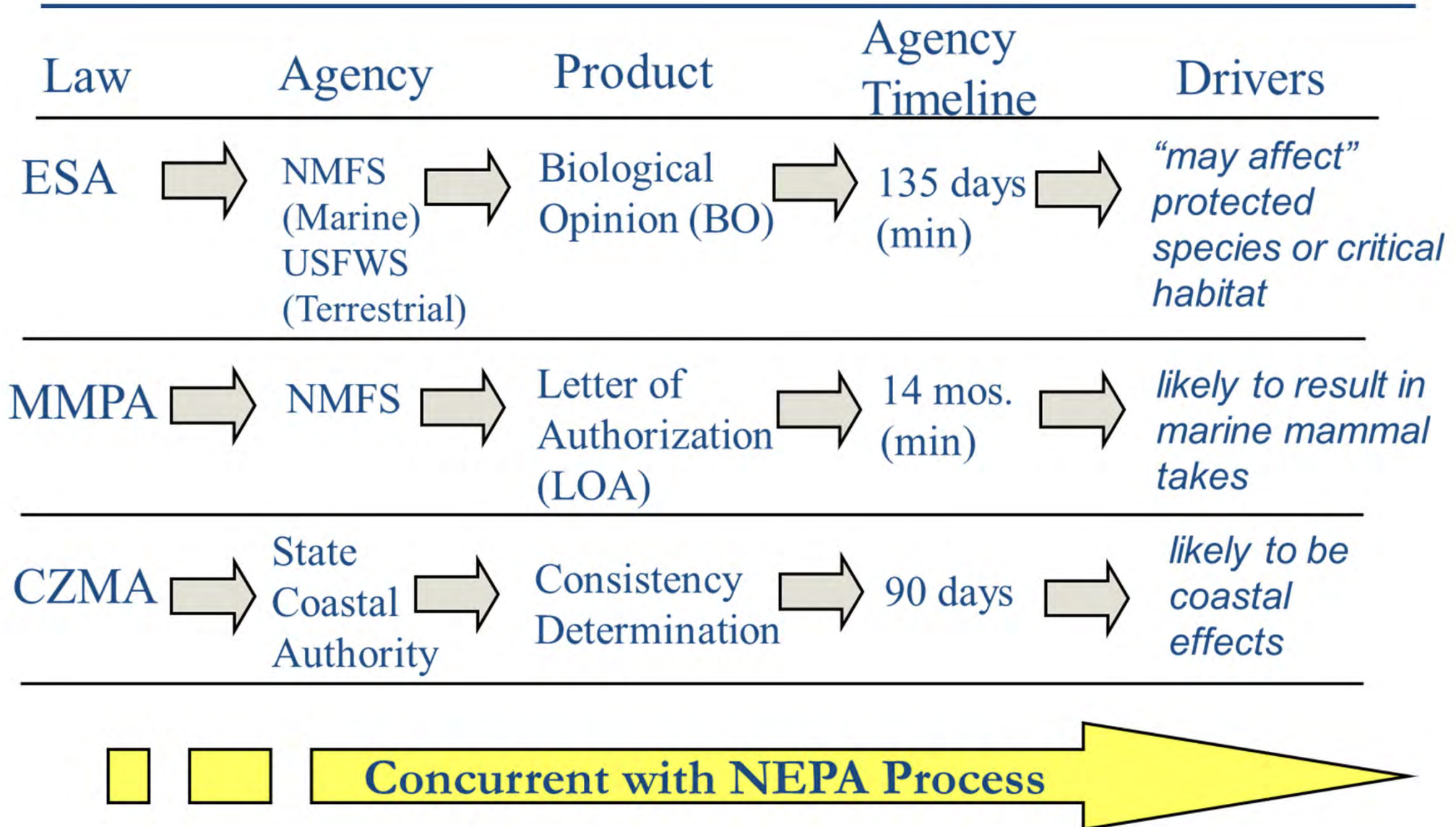


Examples of Other Laws, Statutes, and Executive Orders Typically Integrated with the NEPA Process

- Rivers and Harbors Act of 1899
- The Wild and Scenic Rivers Act
- Clean Water Act
- Safe Drinking Water Act
- Coastal Zone Management Act
- Clean Air Act
- Magnuson Stevens Fishery Conservation and Management Act
- Endangered Species Act
- Bald and Golden Eagle Protection Act
- Migratory Bird Treaty Act
- Protection of Wetlands (EO 11990)
- Floodplain Management (EO 11988)
- Farmland Protection Policy Act
- Protection of Children from Environmental Health Risks and Safety Risks (EO 13045)
- National Historic Preservation Act
- Archaeological Resources Protection Act of 1970
- Native American Graves Protection and Repatriation Act
- American Indian Religious Freedom Act
- Toxic Substances Control Act
- Resource Conservation and Recovery Act
- Comprehensive Environmental Response, Compensation, and Liability Act (Superfund)

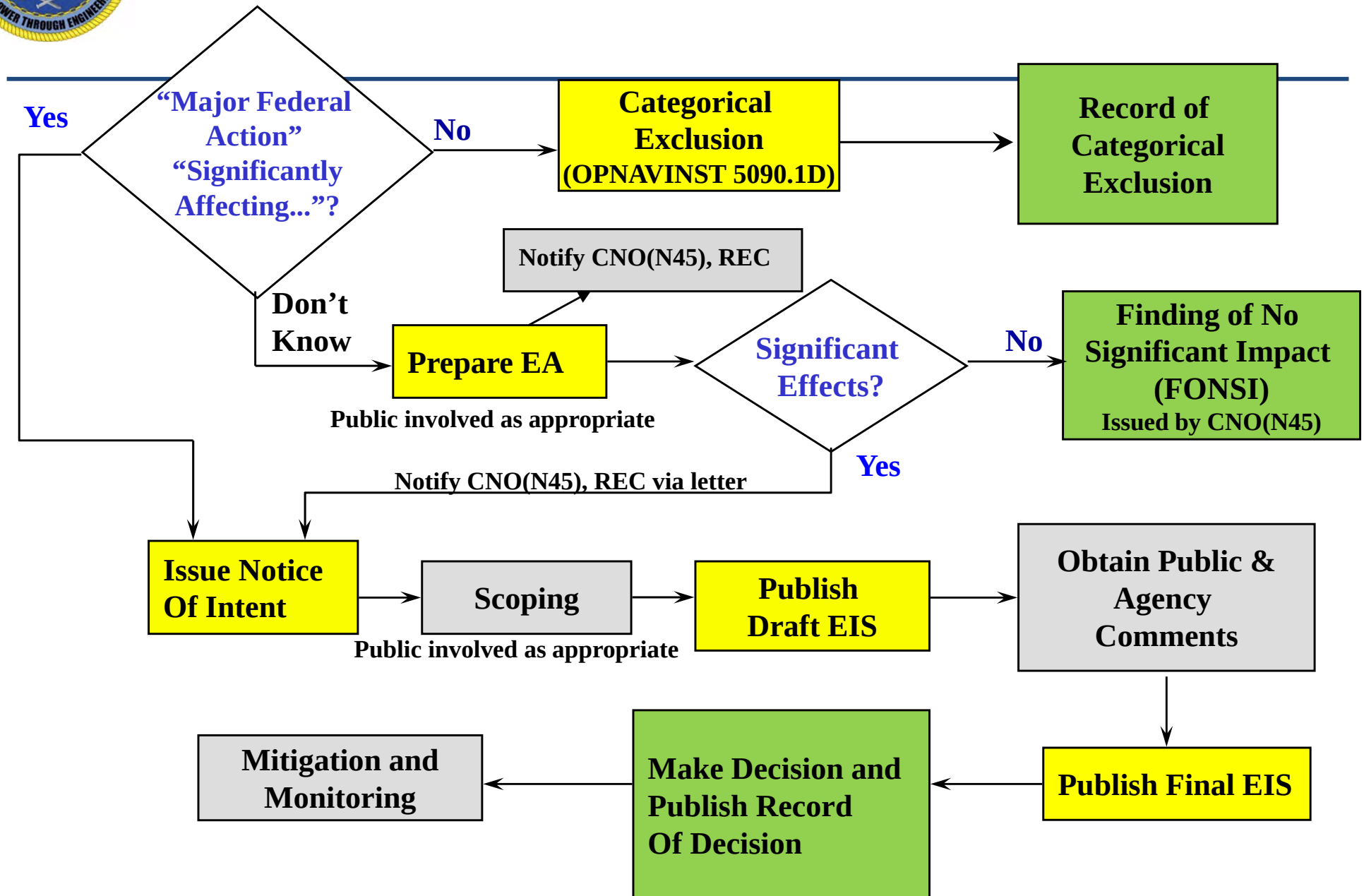


Overlapping Requirements with NEPA





NEPA Process



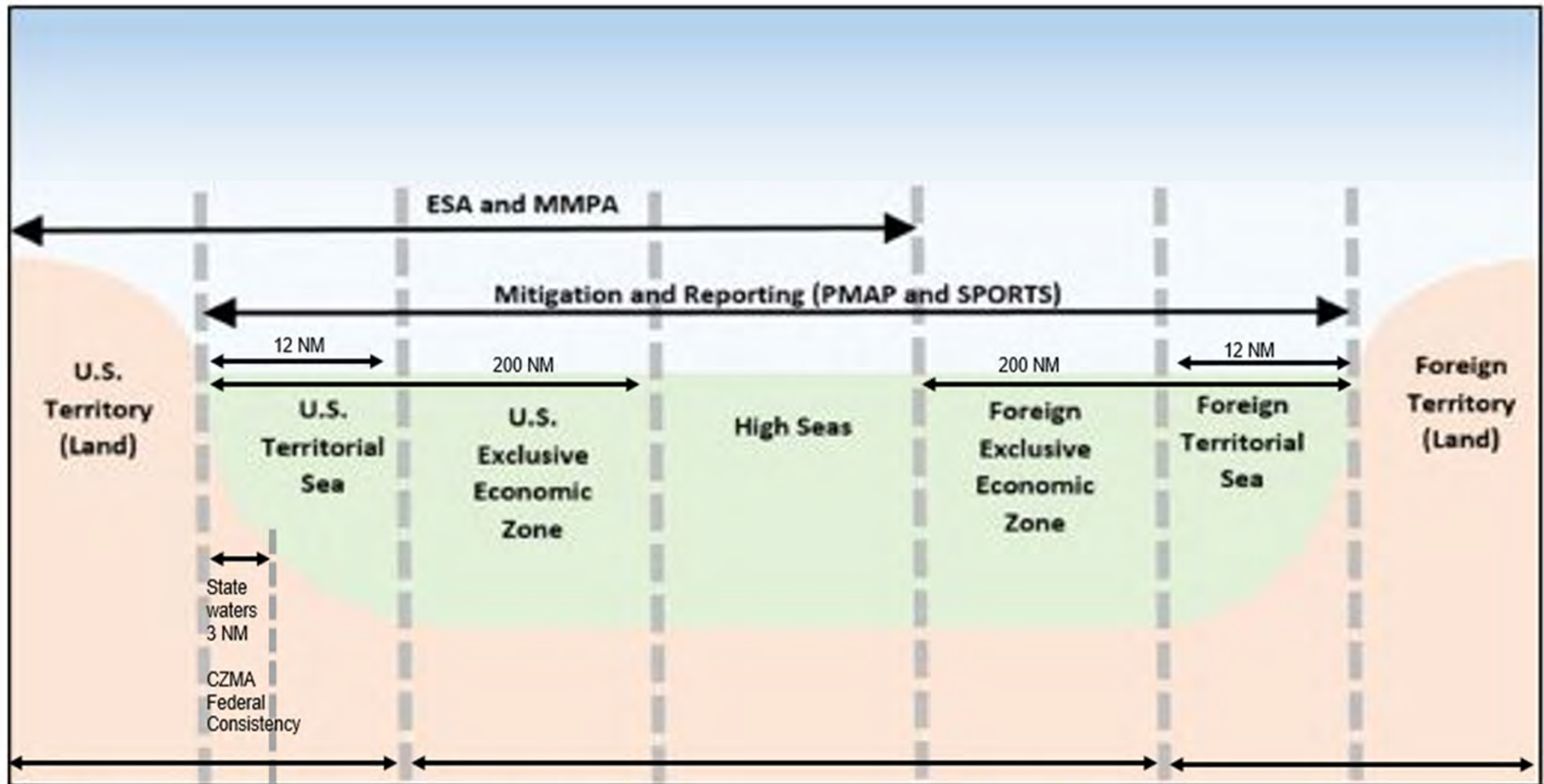


What about Environmental Planning Outside US and Its Territories?

- NEPA does not apply OCONUS except within US territories (e.g. Guam, Puerto Rico, etc.) and their territorial seas within 12-nautical miles of the shoreline
- Executive Order 12114, Environmental Effects Abroad, governs environmental planning abroad
- DoW implementing regulation is 32 CFR 187, Environmental Effects Abroad of Major Department of Defense Actions
- Requires a “NEPA-like analysis” for actions that have potential to significantly harm environment outside US



Overseas At-sea Planning



NEPA:
CATEX / EA / EIS

EO 12114:
Overseas EA
Overseas EIS

EO 12114:
Negative Decision / Environmental Review /
Environmental Study

ESA = Endangered Species Act MMPA = Marine Mammal Protection Act



Overview

- National Environmental Policy Act (NEPA)
- ESOH Laws and Regulations
- DoW Polices and Requirements



Environment, Safety, and Occupational Health (ESOH)

- An acronym that refers to the combination of disciplines that encompass the processes and approaches for addressing laws, regulations, DoW policies, environmental compliance and hazards associated with environmental impacts, system safety (e.g., platforms, systems, system-of-systems, weapons, explosives, software, ordnance, combat systems), occupational safety and health, hazardous materials management, and pollution prevention



ESOH Requirements

- Establish the methods and mechanisms of a process for compliance
 - Local, state, or Federal Agencies may impose civil fines, penalties, or compliance orders resulting in program delays if the compliance process is not properly planned and executed
- Mandate legal compliance requirements and processes
 - **Laws may impose criminal and/or civil penalties** for lack of compliance, which may or may not be applicable to Federal Agencies and individuals
 - Compliance responsibility assigned primarily to the facility and installation managers and maintainers
- For acquisition programs, the PM is responsible for considering ESOH requirements and their effects throughout the system life-cycle when making design decisions

ESOH impacts must be considered as an integral part of any acquisition systems engineering effort



ESOH Law Summary

- AIR POLLUTION CONTROL
 - Clean Air Act (CAA), 42 U.S.C. §§7401-7671q
- HAZARDOUS WASTE
 - Resource Conservation & Recovery Act (RCRA), 42 U.S.C. §§6901-6992k
 - Comprehensive Environmental Response, Compensation & Liability Act (CERCLA), 42 U.S.C. §§ 9601-9675
- WATER POLLUTION CONTROL
 - Federal Water Pollution Control Act (Clean Water Act), 33 U.S.C. §§ 1251-1387
- CHEMICAL PRODUCTION/TRACKING
 - Toxic Substance Control Act (TSCA), 15 U.S.C. §§ 2601-2692
- CHEMICAL EMERGENCY PLANNING & COMMUNITY RIGHT-TO-KNOW
 - Emergency Planning and Community Right-to-Know Act (EPCRA), 42 U.S.C. §§ 11001 - 11050
- ENDANGERED SPECIES PROTECTION
 - Endangered Species Act (ESA) 16 U.S.C. §§1531-1544
 - Marine Mammal Protection Act (MMPA) 16 U.S.C. § 1371-1423h
- WORKPLACE/EMPLOYEE SAFETY
 - Occupational Safety & Health Act (OSH Act), 29 U.S.C. §§ 651-678
- ENVIRONMENTAL POLICY/PLANNING
 - National Environmental Policy Act (NEPA), 42 U.S.C. §§ 4321-4370h
- CULTURAL RESOURCES
 - National Historic Preservation Act, 54 U.S.C. §300101 et seq.



Clean Air Act (CAA)

- Establishes national air quality standards for public's health
- Amended in 1990
- States are responsible for attaining standards
 - State Implementation Plans - govern stationary sources and mobile sources
 - Permits (i.e., boiler operations, paint booths, smokestack emissions, etc.)
 - Reporting
 - Operational impacts
- EPA issues standards to control emissions from aerospace manufacturing & rework and shipbuilding & repair
 - Establishes best technologies for cleaning, painting, etc.
 - Generally, emission reduction through compliant materials and control devices



CAA protects air by reducing or eliminating air pollutants through regulation



Clean Water Act (CWA)

- Major Federal legislation to control water pollution
- Requires National Pollutant Discharge Elimination System (NPDES) permits program focusing on two specific areas which discharge into “waters of the U.S.”
 - Point sources
 - Non-point sources
- Non-point source pollution (biggest challenge)
- Point source examples:
 - Ships
 - Industrial facilities
- Non-point source examples:
 - Highways
 - Parking lots





SCOTUS Curtails Wetlands Protection Under CWA

- *Sackett v EPA* – This dispute involved land a federal appeals court described as a “soggy residential lot” that the landowners wanted to backfill with dirt and rocks so that they could build a house on it
- The lot is near a tributary that feeds into a creek, which itself feeds into Priest Lake, a sufficiently large body of water in Idaho that no one really questions if it is subject to the Clean Water Act
- SCOTUS narrowed CWA jurisdiction to extend only to wetlands that are “as a practical matter indistinguishable from waters of the United States” because they have a “continuous surface connection” with a larger body of water, “making it difficult to determine where the ‘water’ ends and the ‘wetland’ begins”



Resource Conservation and Recovery Act (RCRA)

- Major federal legislation on hazardous waste management, passed in 1976
 - Address growing problem of industrial and municipal waste
 - Set national goals for protecting human health and the environment, conserving energy and natural resources, reducing amount of waste generated, and managing waste in an environmentally-sound manner
- Cradle to grave control of hazardous waste
 - Regulates the generation, management, storage, transport and disposal
 - Manifests created by generator
 - Permits for treatment, storage and disposal
 - **Applies to all shore activities**
- “Aberdeen Three”
 - First time federal employees were tried and convicted in violation of RCRA
 - Improperly storing, treating, and disposing of hazardous waste





Comprehensive Environmental Response Compensation and Liability Act (CERCLA)

- Authorizes federal agency action to respond to the release or threat of release
- Emphasis on cleanup of old/inactive hazardous substance sites
 - Assigned cleanup costs to responsible parties
- Reporting of releases of CERCLA hazardous substances into environment required
 - Report to National Response Center
- Does not cover spills of petroleum, oils, and lubricants (covered by other programs)
- How many bases are superfund sites?



*CERCLA is commonly known as **SUPERFUND***



Toxic Substance Control Act (TSCA)

- Gives the EPA the ability to track the 75,000 industrial chemicals currently produced or imported into the United States. Amended June 2016 to provide EPA with easier way to list and ban chemicals based on risk
 - Can require reporting or testing of those that may pose an environmental or human-health hazard
 - Can ban the manufacture and import of those chemicals that pose an unreasonable risk
- Of concern to DoN and regulated by TSCA:
 - PCBs (Title I)
 - Lead based paint (Title IV)
 - Asbestos (Title II)
 - Radon (Title III)

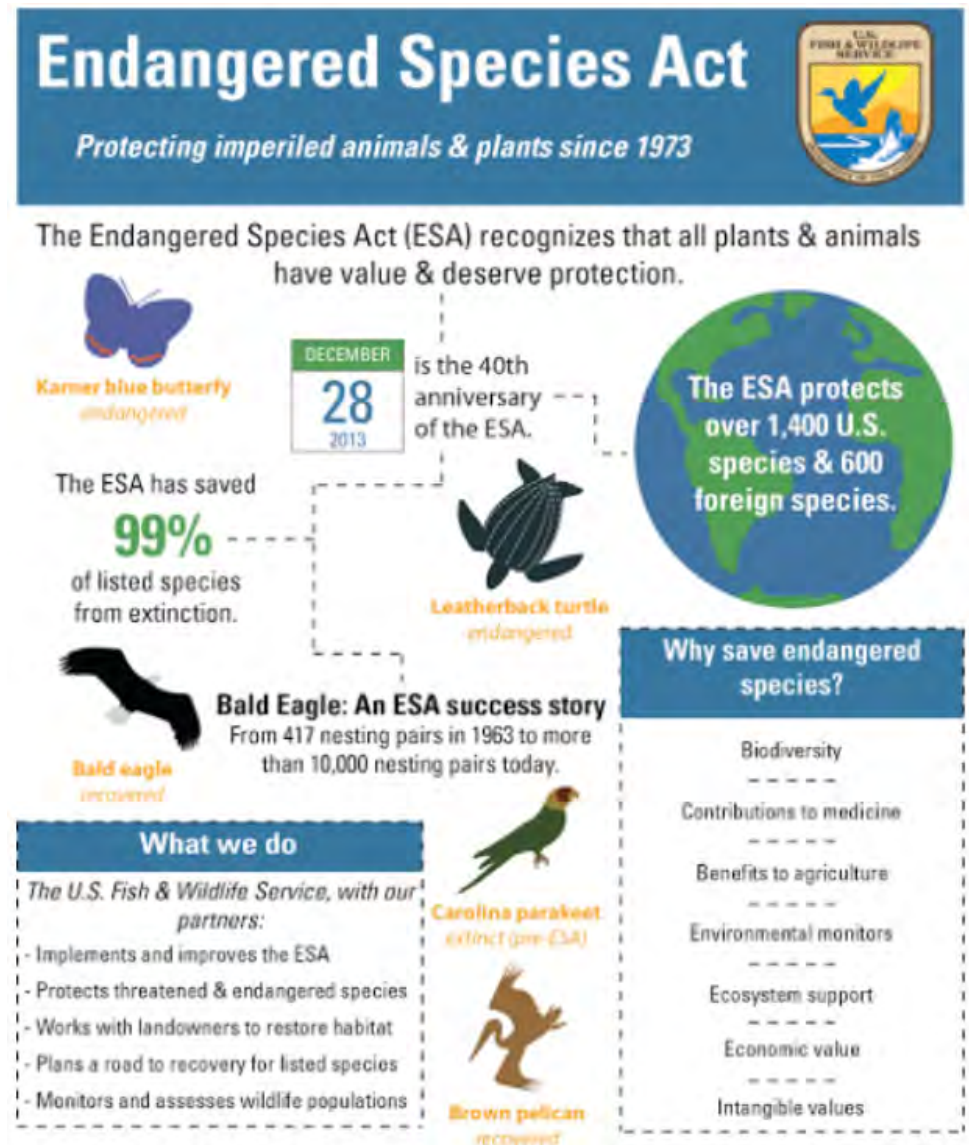


PCBs: Polychlorinated biphenyls



Endangered Species Act (ESA)

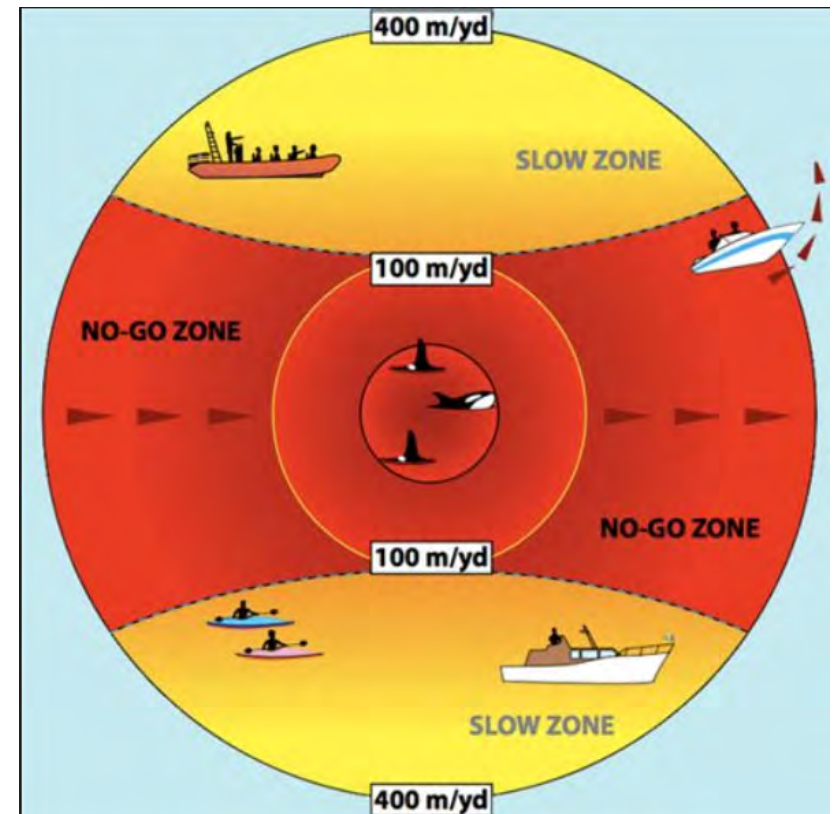
- Law provides broad protections for species of fish, wildlife, and plants that are listed as threatened or endangered - prohibits “taking” of species
- Enacted in 1973 – concern over the decline of various species of fish, wildlife, and plants due to lack of concern and conservation – national symbol going extinct
- Primary purpose – to conserve ecosystems that support endangered and threatened species
- Federal agency duty to consult and affirmatively conserve
- Consultation with FWS/NMFS to ensure compliance





Marine Mammal Protection Act (MMPA)

- Enacted in 1972 – concern over the decline of marine mammal numbers due to human activity
- Primary purpose – to maintain the health and stability of the marine ecosystem
- Some activities often require permission in form of LOA or IHA to “take” or “harass” marine mammals from NOAA as well as NEPA analysis
 - Sonar
 - Sea Trials
 - Weapon Firings



LOA: Letter of Authorization

IHA: Incidental Harassment Authorization

NOAA: National Oceanic and Atmosphere Administration



Cultural Resources

- Few, if any, DoW facilities occupy land devoid of historical interest or cultural resources
- All facilities must comply with a wide range of laws and regulations in this area
- National Historic Preservation Act (NHPA) requires Federal Agency to assess impacts on historic property prior to undertaking action which may affect historic characteristics
- NHPA requires consultation with state and/or Native American tribes for undertakings which may affect historic properties



Building at Pearl Harbor Naval Shipyard is historic. Challenge is modern shipyards, preservation of historic resources



Emergency Planning and Community Right-to-Know Act (EPCRA)

- Requires immediate notification of state and local emergency response planners and the facility's major claimant of releases of hazardous substances over threshold levels
- Provides information to public about potential hazards associated with toxic chemical releases
- Encourages emergency planning at local levels
- Define facility fence line and primary mission
 - Fence line owner (Base Commander) files required EPCRA reports for entire facility
- Releases that result in exposure to personnel solely within facility boundaries do not require notifying state/local agencies





Workplace and Employee Safety

- Occupational Safety & Health Act (OSH Act), 29 U.S.C. §§ 651-678
 - OSHA is common reference
- Addresses worker health and safety
- Federal Government complies with this law by
 - Requirement to provide healthful and safe places and conditions of employment consistent with standards in 29 U.S.C. §655
 - EO 12196 All Federal Agencies must comply with OSHA standards unless Secretary of Labor approves an alternate standard
- “Uniquely military equipment, systems, and operations” - **not** covered under OSHA
 - But DoW workplaces and operations comparable to those of industry in the private sector are covered
- Statute and implementing regulation at 29 CFR Part 1960 cover many different workplaces and operations and contain many record keeping requirements



Overview

- National Environmental Policy Act (NEPA)
- ESOH Laws and Regulations
- DoW Polices and Requirements



DoW Policy

- DoD 5000 series – applies to all ACATs:
 - Prevent, mitigate, or remediate environmental damage caused by acquisition programs
 - Address environmental issues early in development process (throughout life-cycle sustainment)
 - Evaluate alternatives to reduce or eliminate all forms of environmental impact during design, manufacture, test, operation and disposal, at the source whenever feasible
- All DoW personnel (civilian and military), tenants, and Contractors shall comply with all applicable Federal, state, local/internal environmental policies, regulations, and requirements

We are all required to comply with Federal, state, and local environmental laws



Defense Acquisition Program Requirements

- Program Managers must ensure the system can be:
 - produced
 - tested
 - fielded
 - operated
 - trained
 - maintained
 - disposed of
- in compliance with Environment, Safety, and Occupational Health (ESOH) laws, regulations and policy (collectively termed ESOH requirements)



The goal is to safeguard the environment, reduce accidents, and protect human health



Programmatic ESOH Evaluation (PESHE)

- PM must document the program process, the schedule for completing National Environmental Policy Act (NEPA) documentation, and the status of ESOH risk management in a Programmatic ESOH Evaluation (PESHE)
 - PESHE is a management and reporting tool for the PM
 - **Required for ALL ACAT programs during milestone review**
 - Requires analyses in the following areas:
 - Environmental Compliance
 - System Safety and Health
 - Hazardous Materials Management
 - Pollution Prevention Management
 - National Environmental Policy Act (NEPA)



Summary



- What environmental-related document must be updated by program officers before every milestone review?
- Which environmental law allows for discharge into the waters of the US?
- What procedural law established the process for environment review for “major Federal action”?